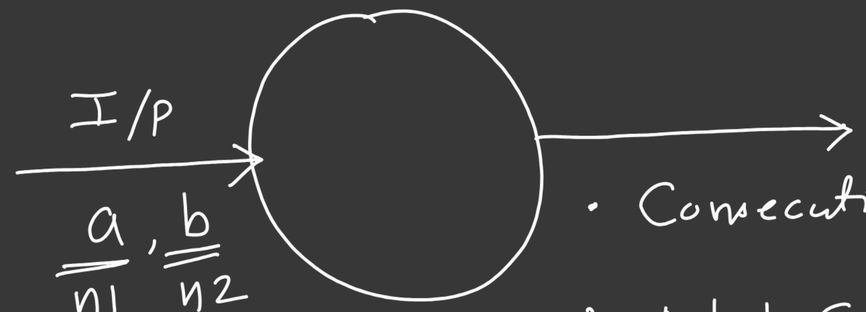
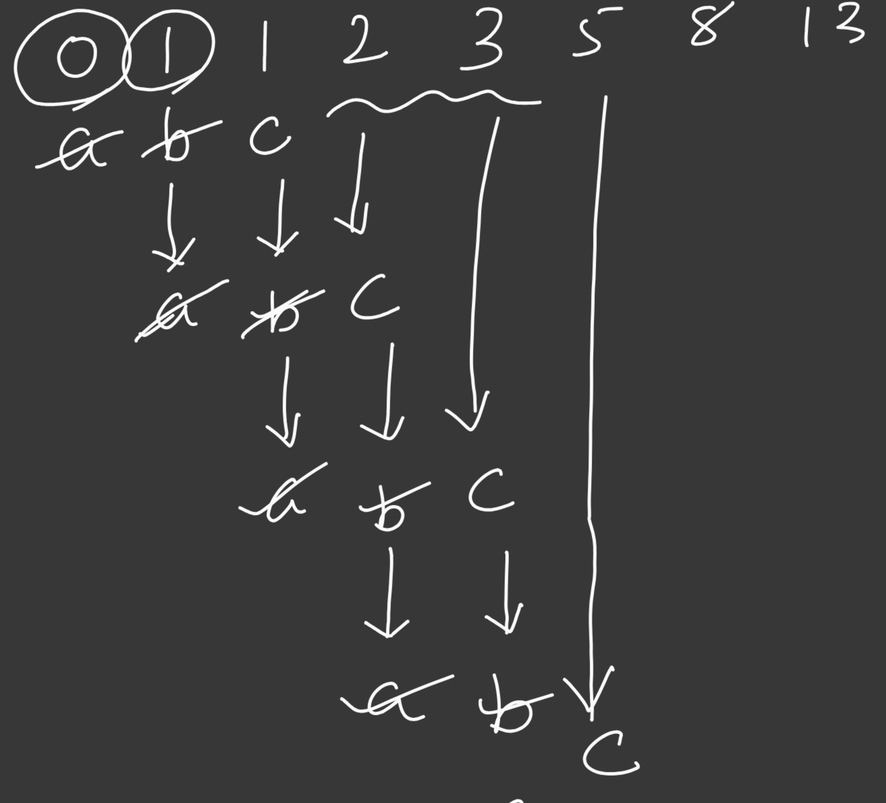
Q:- Checking? Consecutive term in fibonacci series.



- Consecutive
- Not Consecutive.

Assumtn
$n1 < n2$
$a < b$

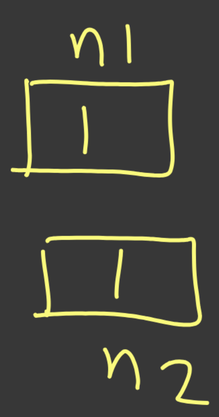


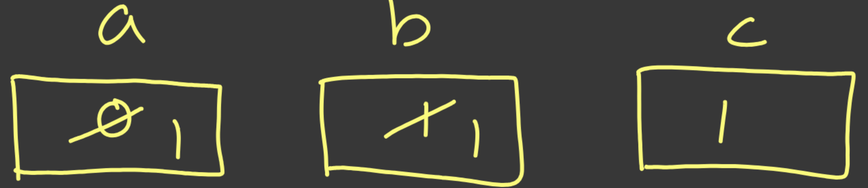
$n1 // a = 2$
$n2 // b = 3$ → Consecutive.

$n1 // a = 5$
$n2 // b = 13$ → Not Consecutive.

$n1 // a = 5$
$n2 // b = 7$ → Not Consecutive.

```
int n1 = sc.nextInt();
int n2 =      "
int a = 0, b = 1, c;
while (a < n1
{
    c = a + b;
    a = b;
    b = c;
}
```

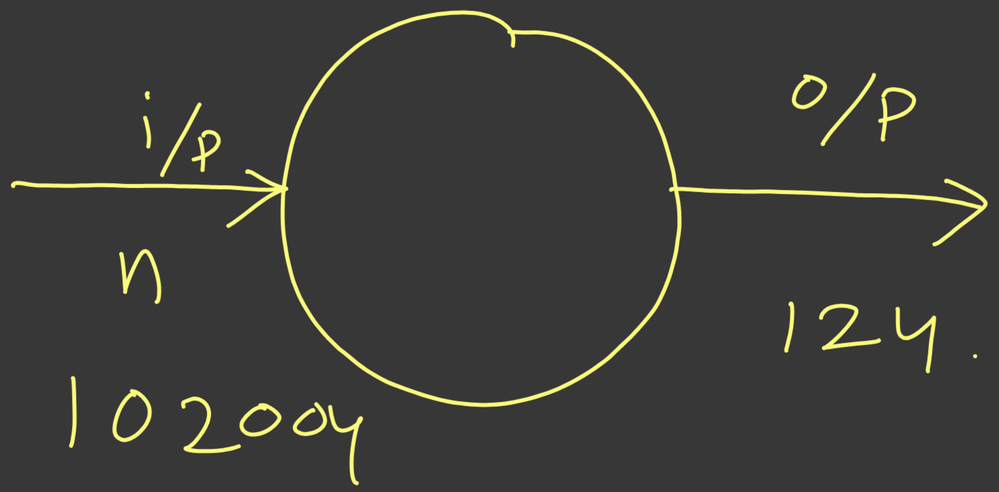





```
if(a == n1 && b == n2)
    s.o.pln("Consecutive");
else
    s.o.pln("Not Consecutive");
```

Q:- Remove Zeros from n.

Let n = 102004 ⟶ 124



e.g

n = 102004 % 10

r = 4

P = 1, num = 0

100 ~~10~~

if (r != 0) 24 + 100 * 1

{

4 + 10 * 2

24 num = num + P*r

~~0~~ 0 + 1 * 4

P = P * 10;

}

10203 `int n = sc.nextInt();`

`int r, num = 0, p = 1;`

`while (n != 0)`

{ ~~+02 +020 +0203~~ != 0

~~0 3~~ `r = n % 10;` 10 % 10 1- 1 % 10

~~2~~ 0 `if (r != 0)` 23 3 + 10 * 2

{ ~~203~~ 3 != 0

`num = num + p * r;` 0 + 1 * 3

`p = p * 10;` 1 * 10

}

`n = n / 10;` 10203/10 = 1020 1020/10 = 102 102/10 = 10 10/10 = 1 1/10 = 0

}

`S.o.pln (num);` = 123

n = 10203

num = 123

23 + 100 * 1 = 123

Q:- Checking a number is a Palindrome number or not.

n = 121

↳ Reverse it Rev = 121

n = = Rev

↳ palindrome.

n = 1023

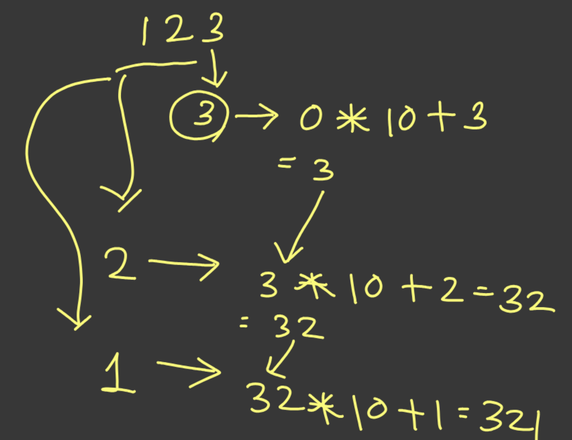
Let n = 123

Rev = 3201

```
int n = sc.nextInt();
int num = n;
int r, rev = 0;
while (n != 0)
{
    r = n % 10;
    rev = rev * 10 + r;
    n = n / 10;
}
```

```
if (num = rev)
{
    s.o.pln("Palindrome");
}
else
{
    s.o.pln("Not Pal");
}
```

rev = rev * 10 + r



Q:-

Write the java stmt to print the multiplication table of 'n' in the given format.

Let n = 5

n		i (1 to 10)		n*i
5	x	1	=	5
5	x	2	=	10
		⋮		
5	x	10	=	50

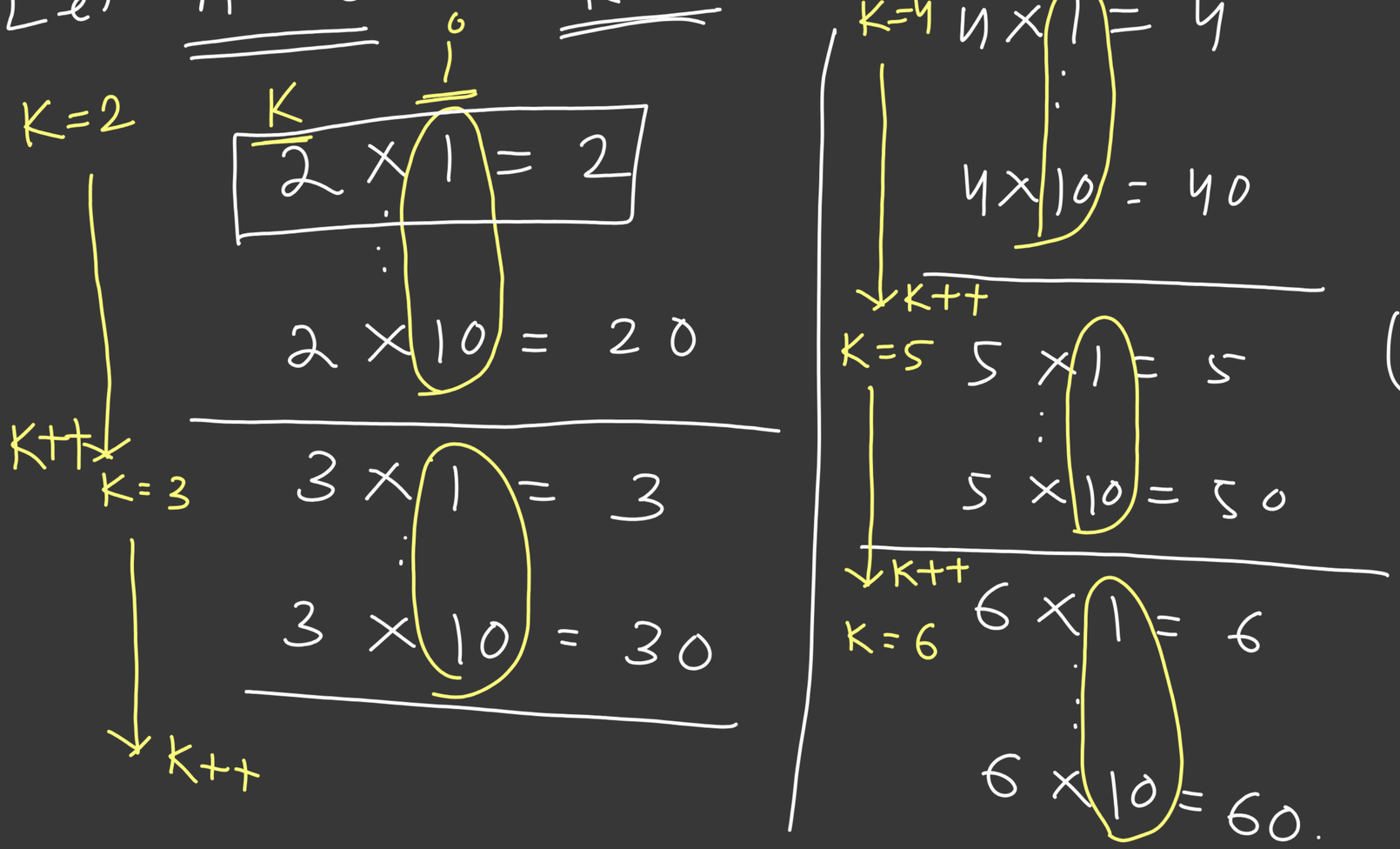
```
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
for(int i=1; i<=10; i++)
{
System.out.println(n+"x"+i+"="+(n*i));
}
```

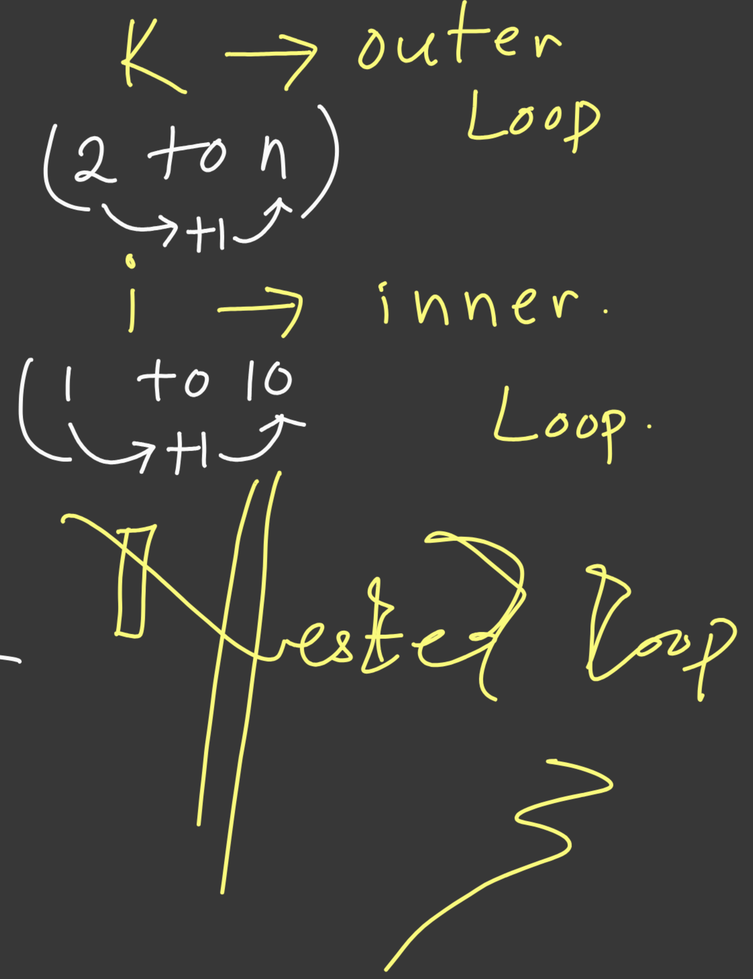
5	x	1	=	5
5	x	2	=	10
		⋮		

Q:— Print the mult^n table from 2 to n.

Let n = 6

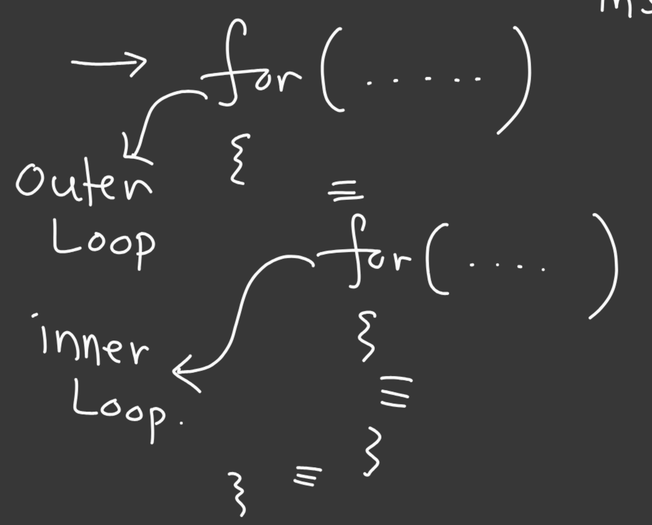
K*i

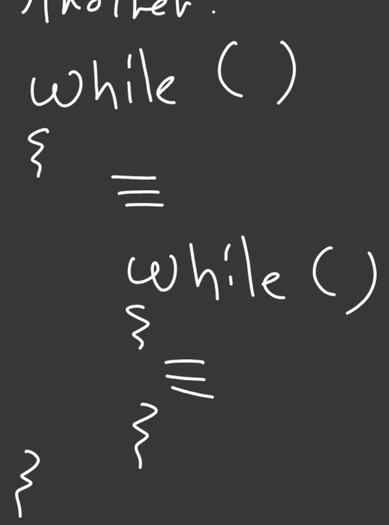


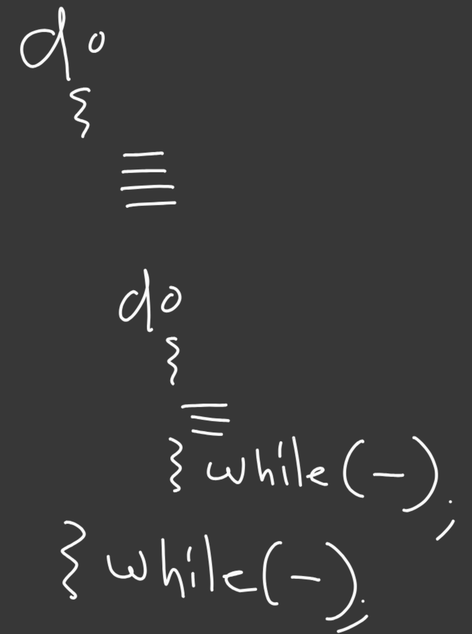


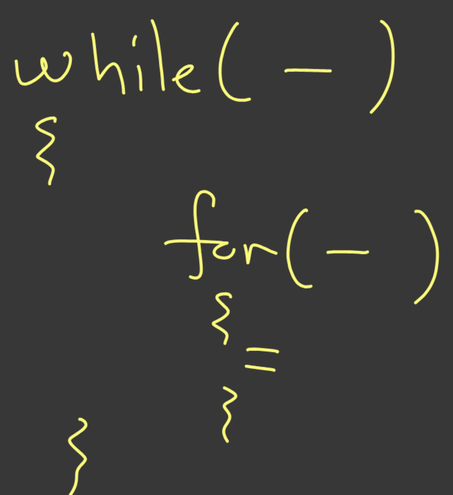
Nested Loop

- One Loop is kept inside Another.
- Any one type of loop can be kept inside Another.

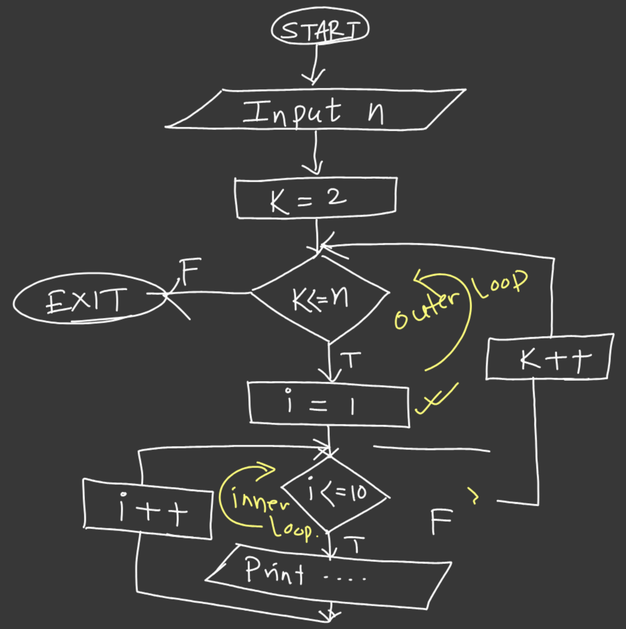








```
int n=sc.nextInt();
int K=2;
while(K<=n)
{
   int i=1;
   while(i<=10)
   {
      s.o.pln(....);
      i++;
   }
   K++;
}
```




```
int n = sc.nextInt();
for(int K=2; K<=n; K++)
{
   for(int i=1; i<=10; i++)
   {
      s.o.pln(K+"x"+i+"="+(K*i));
   }
}
```

Note:- when the control comes from outer loop to inner loop, then inner loop is initialized Again.

2 x 1 = 2
2 x 2 = 4
2 x 10 = 20

3 x 1 = 3